

41. Consider the sets $A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 = 25\}$, $B = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + 9y^2 = 144\}$, $C = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x^2 + y^2 \leq 4\}$ and $D = A \cap B$. The total number of one-one functions from the set D to the set C is :

समुच्चयों $A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 = 25\}$, $B = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + 9y^2 = 144\}$, $C = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x^2 + y^2 \leq 4\}$ और $D = A \cap B$ का विचार कीजिए। समुच्चय D से समुच्चय C के एकैकी फलनों की कुल संख्या है :

- (A) 15120 (B) 19320 (C) 18290 (D) 17160

Ans. D

Sol. $A \equiv x^2 + y^2 = 25$

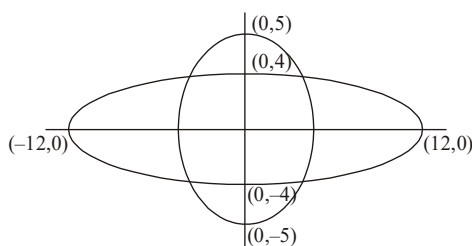
$$B \equiv x^2 + 9y^2 = 144$$

$$C \equiv x^2 + y^2 \leq 4 \quad : x, y \in \mathbb{Z} \times \mathbb{Z}$$

$$\therefore \{(2,0), (0,2), (-2,0), (0,-2)\}$$

$$(1,0) (0,1) (-1,0) (0,-1)$$

$$(1,1) (1,-1) (-1,1) (-1,-1), (0,0)\}$$



$$\therefore \text{number of elements in } n(A \cap B) = 4 \text{ \& } n(C) = 13$$

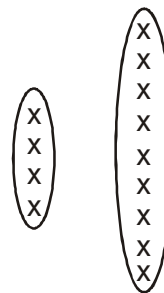
Number of one-one function

$$= {}^{13}C_4 \times 4!$$

$$= \frac{13!}{9!4!} \times 4!$$

$$= 13 \times 12 \times 11 \times 10$$

$$\Rightarrow 17160$$



42. Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 4, 9, 16\}$. Then the number of many-one functions $f: A \rightarrow B$ such that $1 \in f(A)$ is equal to :

माना कि $A = \{1, 2, 3, 4\}$ और $B = \{1, 4, 9, 16\}$ है। तब बहुएकैकी फलनों $f: A \rightarrow B$ की संख्या, इस प्रकार है कि $1 \in f(A)$ है, बराबर है :

- (A) 139 (B) 151 (C) 163 (D) 127

Ans. B

Ans. Total function – (one – one) – $(B - \{1\})$

$$4^4 - 4! - 3^4$$

$$= 151$$

43. Let the range of the function $f(x) = 6 + 16 \cos x \cdot \cos\left(\frac{\pi}{3} - x\right) \cdot \cos\left(\frac{\pi}{3} + x\right) \cdot \sin 3x \cdot \cos 6x$, $x \in \mathbb{R}$ be $[\alpha, \beta]$.

Then the distance of the point (α, β) from the line $3x + 4y + 12 = 0$ is :

माना फलन $f(x) = 6 + 16 \cos x \cdot \cos\left(\frac{\pi}{3} - x\right) \cdot \cos\left(\frac{\pi}{3} + x\right) \cdot \sin 3x \cdot \cos 6x$, $x \in \mathbb{R}$ का परिसर $[\alpha, \beta]$ है। तब बिन्दु $(\alpha,$

$\beta)$ की रेखा $3x + 4y + 12 = 0$ से दूरी है :

(A) 8

(B) 11

(C) 9

(D) 10

Ans. B

Sol. $f(x) = 6 + 16 \cos x \cdot \cos\left(\frac{\pi}{3} - x\right) \cdot \cos\left(\frac{\pi}{3} + x\right) \cdot \sin 3x \cdot \cos 6x$

$$= 6 + 16 \cos x \left[\cos^2 \frac{\pi}{3} - \sin^2 x \right] \sin 3x \cos 6x$$

$$= 6 + 16 \cos x \left[\frac{1}{4} - \sin^2 x \right] \sin 3x \cos 6x$$

$$= 6 + 4 \cos x (1 - 4 \sin^2 x) \sin 3x \cos 6x$$

$$\downarrow$$

$$(1 - \cos^2 x)$$

$$= 6 + 4 \cos x (4 \cos^2 x - 3) \sin 3x \cos 6x$$

$$= 6 + 4[4 \cos^3 x - 3 \cos x] \sin 3x \cos 6x$$

$$= 6 + 4 \cos 3x \sin 3x \cos 6x$$

$$= 6 + \sin 12x$$

$$-1 \leq \sin 12x \leq 1$$

$$5 \leq 6 + \sin 12x \leq 7$$

$$\therefore (\alpha, \beta) \equiv [5, 7]$$

$$d = \frac{|3.5 + 4.7 + 12|}{5}$$

$$d = \frac{|15 + 28 + 12|}{5}$$

$$= 11$$

44. Let $f(x) = \frac{2^{x+2} + 16}{2^{2x+1} + 2^{x+4} + 32}$. Then the value of $8 \left(f\left(\frac{1}{15}\right) + f\left(\frac{2}{15}\right) + \dots + f\left(\frac{59}{15}\right) \right)$ is equal to :

माना कि $f(x) = \frac{2^{x+2} + 16}{2^{2x+1} + 2^{x+4} + 32}$ है। तब $8 \left(f\left(\frac{1}{15}\right) + f\left(\frac{2}{15}\right) + \dots + f\left(\frac{59}{15}\right) \right)$ का मान बराबर है :

(A) 108

(B) 102

(C) 118

(D) 92

Ans. C

Sol. $f(x) = \frac{42^x + 16}{2 \cdot 2^{2x} + 16 \cdot 2^x + 32}$

$$f(x) = \frac{2(2^x + 4)}{2^{2x} + 8 \cdot 2^x + 16}$$

$$f(x) = \frac{2}{2^x + 4}$$

$$f(4-x) = \frac{2^x}{2(2^x + 4)}$$

$$f(x) + f(4-x) = \frac{1}{2}$$

$$\text{So, } f\left(\frac{1}{15}\right) + f\left(\frac{59}{15}\right) = \frac{1}{2}$$

$$\text{Similarly } f\left(\frac{29}{15}\right) + f\left(\frac{31}{15}\right) = \frac{1}{2}$$

$$f\left(\frac{30}{15}\right) = f(2) = \frac{2}{2^2 + 4} = \frac{2}{8} = \frac{1}{4}$$

$$\Rightarrow 8 \left(29 \times \frac{1}{2} + \frac{1}{4} \right)$$

Ans. 118

45. The number of real solution(s) of the equation $x^2 + 3x + 2 = \min\{|x-3|, |x+2|\}$ is :

समीकरण $x^2 + 3x + 2 = \min\{|x-3|, |x+2|\}$ के वास्तविक हल/हलों की संख्या है :

(A) 2

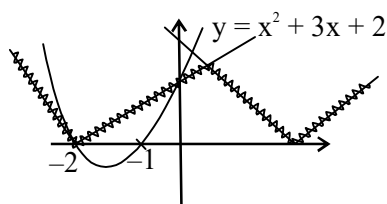
(B) 0

(C) 3

(D) 1

Ans. A

Sol.



46. The function $f: (-\infty, \infty) \rightarrow (-\infty, 1)$, defined by $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$ is :

- (A) Both one-one and onto (B) Neither one-one nor onto
(C) One-one but not onto (D) Onto but not one-one

$f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$ द्वारा परिभाषित फलन $f: (-\infty, \infty) \rightarrow (-\infty, 1)$:

- (A) एकैकी और आच्छादक दोनों है (B) ना तो एकैकी है और ना ही आच्छादक है
(C) एकैकी है परन्तु आच्छादक नहीं (D) आच्छादक है परन्तु एकैकी नहीं है

Ans. C

Sol. $f(x) = \frac{2^{2x} - 1}{2^{2x} + 1} = 1 - \frac{2}{2^{2x} + 1} \in (-1, 1)$

$$f'(x) = \frac{2}{(2^{2x} + 1)^2} \cdot 2^{2x} \cdot 2 \cdot \ln 2 > 0$$

$f(x)$ is one-one but not onto

47. If $f(x) = \frac{2^x}{2^x + \sqrt{2}}$, $x \in \mathbb{R}$, then $\sum_{k=1}^{81} f\left(\frac{k}{82}\right)$ is equal to :

यदि $f(x) = \frac{2^x}{2^x + \sqrt{2}}$, $x \in \mathbb{R}$ है, तो $\sum_{k=1}^{81} f\left(\frac{k}{82}\right)$ बराबर है :

- (A) 82 (B) $\frac{81}{2}$ (C) $81\sqrt{2}$ (D) 41

Ans. B

Sol. $f(x) = \frac{2^x}{2^x + \sqrt{2}}$

$$f(x) + f(1-x) = \frac{2^x}{2^x + \sqrt{2}} + \frac{2^{1-x}}{2^{1-x} + \sqrt{2}}$$

$$= \frac{2^x}{2^x + \sqrt{2}} + \frac{2}{2 + \sqrt{2}2^x} = \frac{2^x + \sqrt{2}}{2^x + \sqrt{2}} = 1$$

$$\text{Now, } \sum_{k=1}^{81} f\left(\frac{k}{82}\right) = f\left(\frac{1}{82}\right) + f\left(\frac{2}{82}\right) + \dots + f\left(\frac{81}{82}\right)$$

$$= f\left(\frac{1}{82}\right) + f\left(\frac{1}{82}\right) + \dots + f\left(1 - \frac{2}{82}\right) + f\left(1 - \frac{1}{82}\right)$$

$$\left[f\left(\frac{1}{82}\right) + f\left(1 - \frac{1}{82}\right) \right] + \left[f\left(\frac{2}{82}\right) + f\left(1 - \frac{2}{82}\right) \right] + \dots + 40 \text{ cases} + f\left(\frac{41}{82}\right)$$

$$(1 + 1 + \dots + 1) 40 \text{ times} + \frac{2^{\frac{1}{2}}}{2^{\frac{1}{2}} + 2^{\frac{1}{2}}}$$

$$40 + \frac{1}{2} = \frac{81}{2}$$

48. Let $f: \mathbb{R} - \{0\} \rightarrow (-\infty, 1)$ be a polynomial of degree 2, satisfying $f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$. If $f(K) = -2K$, then the sum of squares of all possible values of K is :

माना $f: \mathbb{R} - \{0\} \rightarrow (-\infty, 1)$ घात 2 का एक बहुपद है, जो कि $f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ को संतुष्ट करता है। यदि

$f(K) = -2K$, तो K के सभी संभव मानों के वर्गों का योग है :

- (A) 9 (B) 7 (C) 1 (D) 6

Ans. D

Sol. $f(x) = (1 - x^2)$
 $f(k) = (1 - k^2) = -2k$

$$k^2 - 2k - 1 = 0 \begin{cases} \alpha \\ \beta \end{cases}$$

$$(\alpha^2 + \beta^2) = (\alpha + \beta)^2 - 2\alpha\beta$$

$$\Rightarrow (2)^2 - 2(-1)$$

$$\Rightarrow 4 + 2 = 6$$

49. Let $f: [0, 3] \rightarrow A$ be defined by $f(x) = 2x^3 - 15x^2 + 36x + 7$ and $g: [0, \infty) \rightarrow B$ be defined by $g(x) = \frac{x^{2025}}{x^{2025} + 1}$. If both the functions are onto and $S = \{x \in \mathbb{Z} : x \in A \text{ or } x \in B\}$, then $n(S)$ is equal to :

माना कि $f: [0, 3] \rightarrow A$, $f(x) = 2x^3 - 15x^2 + 36x + 7$ द्वारा परिभाषित होता है और $g: [0, \infty) \rightarrow B$, $g(x) = \frac{x^{2025}}{x^{2025} + 1}$ द्वारा

परिभाषित होता है। यदि दोनों फलन आच्छादक हैं और $S = \{x \in \mathbb{Z} : x \in A \text{ या } x \in B\}$ है, तो $n(S)$ बराबर है :

- (A) 30 (B) 31 (C) 36 (D) 29

Ans. A

Sol. $S = \{x \in \mathbb{Z} (\text{Integers}) : x \in A \text{ or } x \in B\}$

$\Rightarrow$ All Integer values in set A & set B.

(i) $f: [0, 3] \rightarrow A$

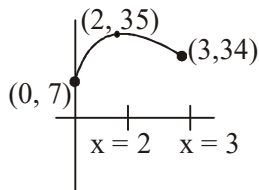
$$f(x) = 2x^3 - 15x^2 + 36x + 7$$

$$f'(x) = 6x^2 - 30x + 36$$

$$\Rightarrow 6(x^2 - 5x + 6)$$

$$\Rightarrow 6(x-3)(x-2)$$

$$\begin{array}{c} + \quad - \quad + \\ \uparrow \quad 2 \quad \downarrow \quad 3 \quad \uparrow \end{array}$$



$$f(2) = 2(8) - 15(4) + 36(2) + 7$$

$$\Rightarrow 16 - 60 + 72 + 7$$

$$\Rightarrow (28 + 7) = 35$$

$$f(3) = 2(27) - 15(9) + 36(3) + 7$$

$$\Rightarrow 54 - 135 + 108 + 7$$

$$\Rightarrow 34$$

$$A \Rightarrow [7, 35] \Rightarrow \text{Total integers} \rightarrow 29$$

$$(ii) g : [0, \infty) \rightarrow B$$

$$g(x) = \left(\frac{x^{2025}}{x^{2025} + 1} \right)$$

$$g(x) = 1 - \frac{1}{(x^{2025} + 1)}$$

$$B \Rightarrow [0, 1) \Rightarrow \text{Total integer} = 1$$

$$n(S) = A + B \Rightarrow (29 + 1) = 30 \text{ Integers}$$

50. Let $A = \{1, 3, 7, 9, 11\}$ and $B = \{2, 4, 5, 7, 8, 10, 12\}$. Then the total number of one-one maps $f: A \rightarrow B$, such that $f(1) + f(3) = 14$, is :

माना $A = \{1, 3, 7, 9, 11\}$ तथा $B = \{2, 4, 5, 7, 8, 10, 12\}$ हैं। तो एकैकी प्रतिचित्रों (maps) $f: A \rightarrow B$, जिनके लिए $f(1) + f(3) = 14$ है, की कुल संख्या है :

- (A) 180 (B) 120 (C) 480 (D) 240

Ans. D

Sol. $A = \{1, 3, 7, 9, 11\}$

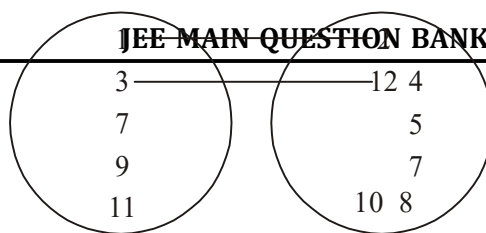
$$B = \{2, 4, 5, 7, 8, 10, 12\}$$

$$f(1) + f(3) = 14$$

$$(i) 2 + 12$$

$$(ii) 4 + 10$$

$$2 \times (2 \times 5 \times 4 \times 3) = 240$$



51. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as :

$$f(x) = |x-1| \text{ and } g(x) = \begin{cases} e^x, & x \geq 0 \\ x+1, & x \leq 0 \end{cases}$$

Then the function $f(g(x))$ is :

(A) Neither one-one nor onto

(B) Onto but not one-one

(C) One-one but not onto

(D) Both one-one and onto

माना $f, g : \mathbb{R} \rightarrow \mathbb{R}$,

$$f(x) = |x-1| \text{ तथा } g(x) = \begin{cases} e^x, & x \geq 0 \\ x+1, & x \leq 0 \end{cases}$$

द्वारा परिभाषित हैं। तो फलन $f(g(x))$:

(A) न तो एकैकी न ही आच्छादक है

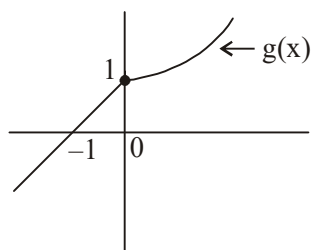
(B) आच्छादक है परन्तु एकैकी नहीं है

(C) एकैकी है परन्तु आच्छादक नहीं है

(D) एकैकी तथा आच्छादक दोनों है

Ans. A

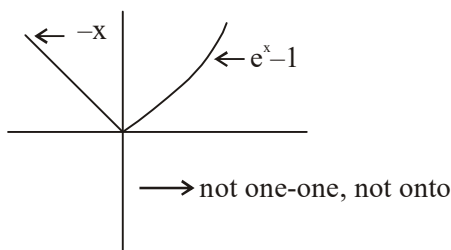
Sol.
$$f(x) = \begin{cases} x-1 & x \geq 1 \\ 1-x & x < 1 \end{cases}$$



$$g(x) = \begin{cases} e^x & ; x \geq 0 \\ x+1 & ; x \leq 0 \end{cases}$$

$$f(g(x)) = \begin{cases} g(x) - 1, & g(x) \geq 1 \\ 1 - g(x), & g(x) < 1 \end{cases}$$

$$= \begin{cases} e^x - 1 & x \geq 0 \\ -x & x < 0 \end{cases}$$



52. The function $f: \mathbb{N} - \{1\} \rightarrow \mathbb{N}$; is defined by $f(n)$ = the highest prime factor of n , is :

- (A) neither one-one nor onto (B) onto only
(C) one-one only (D) both one-one and onto

फलन $f: \mathbb{N} - \{1\} \rightarrow \mathbb{N}$; $f(n)$ = पूर्णांक n का सबसे बड़ा अभाज्य गुणनखंड, द्वारा परिभाषित है। तो f :

- (A) न तो एकैकी है न ही आच्छादक है (B) केवल आच्छादक है
(C) केवल एकैकी है (D) एकैकी और आच्छादक दोनों है

Ans. A

Sol. Largest prime factors of two different natural numbers can be same so it is many one.

For example $f(6) = 3, f(12) = 3$

$f(10) = 5, f(15) = 5$

As range of $f(x)$ is set of prime number so range of $f(x)$ is proper subset of co-domain so f is into.

53. If $f(x) = \begin{cases} 2 + 2x, & -1 \leq x < 0 \\ 1 - \frac{x}{3}, & 0 \leq x \leq 3 \end{cases}$; $g(x) = \begin{cases} -x, & -3 \leq x \leq 0 \\ x, & 0 < x \leq 1 \end{cases}$, then range of $(f \circ g)(x)$ is :

यदि $f(x) = \begin{cases} 2 + 2x, & -1 \leq x < 0 \\ 1 - \frac{x}{3}, & 0 \leq x \leq 3 \end{cases}$; $g(x) = \begin{cases} -x, & -3 \leq x \leq 0 \\ x, & 0 < x \leq 1 \end{cases}$, तो $(f \circ g)(x)$ का परिसर है :

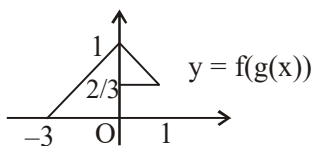
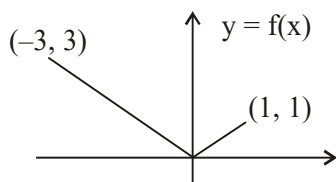
- (A) $[0, 1)$ (B) $[0, 3)$ (C) $(0, 1]$ (D) $[0, 1]$

Ans. D

Sol. $f(g(x)) = \begin{cases} 2+2g(x) & , \quad -3 \leq g(x) < 0 \quad \text{--- (1)} \\ 1 - \frac{g(x)}{3} & , \quad 0 \leq g(x) \leq 3 \quad \text{--- (2)} \end{cases}$

By (1) $x \in \phi$

And by (2) $x \in [-3, 0]$ and $x \in [0, 1]$



Range of $f(g(x))$ is $[0, 1]$

54. If $f(x) = \frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$ and $(f \circ f)(x) = g(x)$, where $g : \mathbb{R} - \left\{\frac{2}{3}\right\} \rightarrow \mathbb{R} - \left\{\frac{2}{3}\right\}$, then $(g \circ g \circ g)(4)$ is equal to :

यदि $f(x) = \frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$ तथा $(f \circ f)(x) = g(x)$ है, जहाँ $g : \mathbb{R} - \left\{\frac{2}{3}\right\} \rightarrow \mathbb{R} - \left\{\frac{2}{3}\right\}$ है, तो $(g \circ g \circ g)(4)$ बराबर है:

- (A) -4 (B) $-\frac{19}{20}$ (C) $\frac{19}{20}$ (D) 4

Ans. D

Sol. $g(x) = f\left(\frac{4x+3}{6x-4}\right) = \frac{4\left(\frac{4x+3}{6x-4}\right) + 3}{6\left(\frac{4x+3}{6x-4}\right) - 4} = \frac{16x+12+18x-12}{24x+18-24x+16} = \frac{34x}{34} = x$

$g \circ g \circ g(4) = g(g(4)) = g(4) = 4$

55. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} \log_e x & , \quad x > 0 \\ e^{-x} & , \quad x \leq 0 \end{cases}$ and $g(x) = \begin{cases} x & , \quad x \geq 0 \\ e^x & , \quad x < 0 \end{cases}$. Then

$g \circ f : \mathbb{R} \rightarrow \mathbb{R}$ is :

- (A) both one-one and onto (B) onto but not one-one
(C) one-one but not onto (D) neither one-one nor onto

माना $f : \mathbb{R} \rightarrow \mathbb{R}$ तथा $g : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \begin{cases} \log_e x & , \quad x > 0 \\ e^{-x} & , \quad x \leq 0 \end{cases}$ तथा $g(x) = \begin{cases} x & , \quad x \geq 0 \\ e^x & , \quad x < 0 \end{cases}$ द्वारा परिभाषित है। तो

$g \circ f : \mathbb{R} \rightarrow \mathbb{R}$:

- (A) एकैकी एवं आच्छादक दोनों है
(C) एकैकी है परन्तु आच्छादक नहीं है

- (B) आच्छादक है परन्तु एकैकी नहीं है
(D) न तो एकैकी है, न ही आच्छादक है

Ans. D

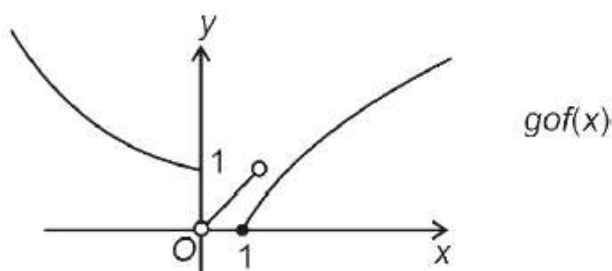
Sol. Given $f(x) = \begin{cases} e^{-x} & ; x \leq 0 \\ \ln x & ; x > 0 \end{cases}$

and $g(x) = \begin{cases} x & ; x \geq 0 \\ e^x & ; x < 0 \end{cases}$

then $g \circ f(x)$
 $= g(f(x))$

$= \begin{cases} f(x) & ; f(x) \geq 0 \\ e^{f(x)} & ; f(x) < 0 \end{cases}$

$= \begin{cases} e^{-x} & ; x \leq 0 \\ x & ; 0 < x < 1 \\ \ln x & ; x \geq 1 \end{cases}$



$\therefore f(x)$ is neither one-one nor onto

56. Let $A = \{x \in \mathbb{R} : [x+3] + [x+4] \leq 3\}$, $B = \left\{x \in \mathbb{R} : 3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^r} \right)^{x-3} < 3^{-3x}\right\}$, where $[t]$ denotes greatest integer function. Then :

माना $A = \{x \in \mathbb{R} : [x+3] + [x+4] \leq 3\}$, $B = \left\{x \in \mathbb{R} : 3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^r} \right)^{x-3} < 3^{-3x}\right\}$ है, जहाँ $[t]$ महत्तम पूर्णांक फलन है। तब

:

- (A) $A = B$ (B) $B \subset A, A \neq B$ (C) $A \cap B = \phi$ (D) $A \subset B, A \neq B$

Ans. A

Sol. $[x] + 3 + [x] + 4 \leq 3$

$$2[x] \leq -4$$

$$[x] \leq -2 \Rightarrow x \in (-\infty, -1). \quad \text{_____A}$$

$$3^x \left(\frac{3 \cdot \frac{1}{10}}{1 - \frac{1}{10}} \right)^{x-3} < 3^{-3x}$$

$$27 < 3^{-3x}$$

$$-3x > +3$$

$$x < -1$$

$$A = B$$

$$\text{_____B}$$

57. If $f(x) = \frac{(\tan 1^\circ)x + \log_e(123)}{x \log_e(1234) - (\tan 1^\circ)}$, $x > 0$, then the least value of $f(f(x)) + f\left(f\left(\frac{4}{x}\right)\right)$ is :

यदि $f(x) = \frac{(\tan 1^\circ)x + \log_e(123)}{x \log_e(1234) - (\tan 1^\circ)}$, $x > 0$ है, तो $f(f(x)) + f\left(f\left(\frac{4}{x}\right)\right)$ का निम्नतम मान है :

(A) 2

(B) 8

(C) 4

(D) 0

Ans. C

Sol. $f(x) = \frac{(\tan 1)x + \log 123}{x \log 1234 - \tan 1}$

Let $A = \tan 1, B = \log 123, C = \log 1234$

$$f(x) = \frac{Ax + B}{xC - A}$$

$$f(f(x)) = \frac{A\left(\frac{Ax+B}{xC-A}\right) + B}{C\left(\frac{Ax+B}{xC-A}\right) - A}$$

$$= \frac{A^2x + AB + xBC - AB}{ACx + BC - ACx + A^2}$$

$$= \frac{x(A^2 + BC)}{(A^2 + BC)} = x$$

$$f(f(x)) = x$$

$$f\left(f\left(\frac{4}{x}\right)\right) = \frac{4}{x}$$

$$f(f(x)) + f\left(f\left(\frac{4}{x}\right)\right)$$

$$AM \geq GM$$

$$x + \frac{4}{x} \geq 4$$

58. The equation $x^2 - 4x + [x] + 3 = x[x]$, where $[x]$ denotes the greatest integer function, has :

- (A) no solution (B) a unique solution in $(-\infty, \infty)$
 (C) a unique solution in $(-\infty, 1)$ (D) exactly two solution in $(-\infty, \infty)$

समीकरण $x^2 - 4x + [x] + 3 = x[x]$, जहाँ $[x]$ महत्तम पूर्णांक फलन है :

- (A) का कोई हल नहीं है (B) का $(-\infty, \infty)$ में ठीक एक हल है
 (C) का $(-\infty, 1)$ में ठीक एक हल है (D) का $(-\infty, \infty)$ में ठीक दो हल है

Ans. B

Sol. $x^2 - 4x + [x] + 3 = x[x]$
 $\Rightarrow x^2 - 4x + 3 = x[x] - [x]$
 $\Rightarrow (x-1)(x-3) = [x] \cdot (x-1)$
 $\Rightarrow x = 1$ or $x - 3 = [x]$
 $\Rightarrow x - [x] = 3$
 $\Rightarrow \{x\} = 3$ (Not Possible)
 Only one solution $x = 1$ in $(-\infty, \infty)$

59. The number of functions $f : \{1, 2, 3, 4\} \rightarrow \{a \in \mathbb{Z} : |a| \leq 8\}$ satisfying $f(n) + \frac{1}{n}f(n+1) = 1, \forall n \in \{1, 2, 3\}$ is

$$f(n) + \frac{1}{n}f(n+1) = 1, \forall n \in \{1, 2, 3\} \text{ को संतुष्ट करने वाले फलनों } f : \{1, 2, 3, 4\} \rightarrow \{a \in \mathbb{Z} : |a| \leq 8\}$$

की संख्या है

- (A) 3 (B) 2 (C) 4 (D) 1

Ans. B

Sol. $nf(n) + f(n+1) = n$
 $f(1) + f(2) = 1$
 $2f(2) + f(3) = 2$
 $3f(3) + f(4) = 3$
 $2 \cdot (1 - f(1)) + f(3) = 2$

$$f(3) = 2f(1)$$

$$3.2f(1) + f(4) = 3$$

$$f(4) = 3 - 6f(1) \leq 8$$

$$-8 \leq f(4) \leq 8$$

$$-8 \leq 3 - 6f(1) \leq 8$$

$$-11 \leq -6f(1) \leq 5$$

$$-\frac{5}{6} \leq f(1) \leq \frac{11}{6}$$

$$f(1) = 0, 1$$

$$\text{C-I: } f(1) = 0, \Rightarrow f(2) = 1$$

$$f(3) = 0, f(4) = 3$$

$$\text{C-II: } f(1) = 1, \Rightarrow f(2) = 0$$

$$\Rightarrow f(3) = 2, f(4) = -3$$

so '2' such functions.

60. Number of functions $f: \{1, 2, \dots, 100\} \rightarrow \{0, 1\}$, that assign 1 to exactly one of the positive integers less than or equal to 98, is equal to _____.

फलनों $f: \{1, 2, \dots, 100\} \rightarrow \{0, 1\}$, जो 98 से छोटे या उसके बराबर धनात्मक पूर्णाकों में से ठीक एक को 1 पर निरूपित करते हैं, की संख्या है _____.

Ans. 392

Sol. No. of ways to connect one of

$$\{1, 2, 3, \dots, 98\} \text{ to } 1 : {}^{98}C_1 = 98$$

$$\text{No. of ways to connect 99 and 100} = 2 \times 2$$

$$\text{Required ans } 98 \times 4 = 392$$